

Pre-lab Questions

1. What is the difference between decoder and data distributor?

A decoder identifies which output corresponds to the binary input while a data distributor selects an output based on the input and also sends data through the selected line

2. How many output lines would a 5-to-32 decoder have?

Outputs= $2^n=2^5=32$

A 5-to-32 decoder has 32 output lines.

3. Explain how a decoder can be used to drive a 7-segment display.

A decoder can translate binary or BCD inputs into signals that can light up specific sections of a 7-segment display. (a-g)

Post lab Questions

1. How does the choice of modeling style affect the Verilog code for the decoders?

The modeling style determines how the decoder's behavior is described. Different styles can affect the readability and level of detail of the verilog code

2. What is the purpose of enable inputs in a decoder?

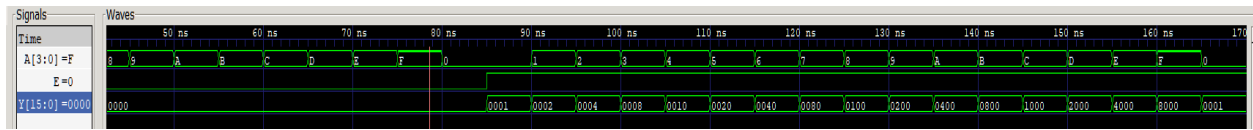
The enable input controls whether the decoder is active. When enabled, the decoder operates normally and activates one output based on the input. When disabled, all outputs are inactive. This allows decoders to be combined or controlled selectively in larger systems.

3. What are some real-world applications of decoders?

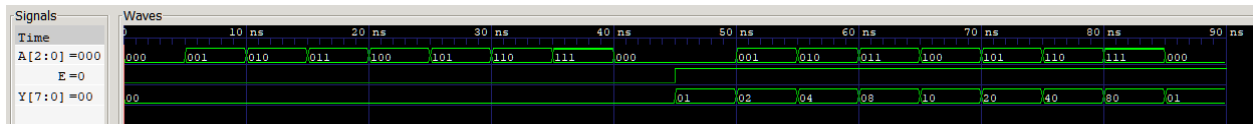
Decoders are used in memory address selection, 7-segment displays, CPU instruction decoding, and communication systems. They help select specific devices, convert binary data to readable output, and control signals in digital circuits.

Decimal Number	Input Lines				Output Lines						
	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1
10	1	0	1	0	1	1	1	0	1	1	1
11	1	0	1	1	0	0	1	1	1	1	1
12	1	1	0	0	1	0	0	1	1	1	0
13	1	1	0	1	0	1	1	1	1	0	1
14	1	1	1	0	1	0	0	1	1	1	1
15	1	1	1	1	1	0	0	0	1	1	1

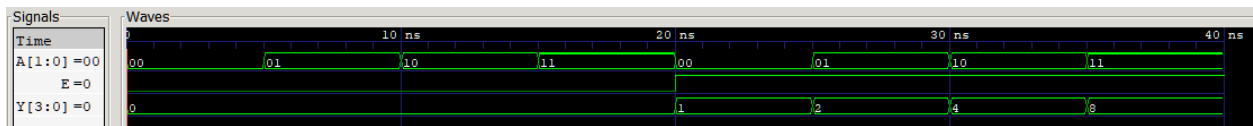
4:16 Decoder



3:8 Decoder



2:4 Decoder



Verilog Code: <https://github.com/heapdats/COE180.1>